Mindmatrix Project no 90

Android App Development using GenAI - Karunada-Kala Source (National Pride)

Cultural Discovery & Artisan Platform — Android Application

Full Project Documentation

Name : Roshan Attar
USN : 2LB22AI017

College: Biluru Gurubasava Mahaswamiji Institute of Technology Mudhol

Email id:roshanattar2004@gmail.com

Project Name: Android App Development using GenAI - Karunada-Kala Source (National Pride)

Platform: Android (GenAI-Powered)

Target Audience: Students, Tourists, Art Enthusiasts and Rural Artisans across Karnataka

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1. Project Overview

Karunada-Kala Source is an advanced Android-based application developed to digitally preserve, promote, and connect Karnataka's traditional art ecosystem with modern users. The term "Karunada" refers to Karnataka, and "Kala" represents art, symbolizing the cultural identity of the region.

The application acts as a **centralized cultural discovery platform** that integrates multiple functionalities such as art exploration, artisan discovery through maps, cultural event tracking, workshop booking, and AI-powered assistance. Unlike traditional platforms, this application not only displays cultural content but also enables direct interaction between users and artisans.

The system is designed to function as a bridge between rural artisans and urban or global audiences by leveraging modern technologies like cloud databases, geolocation services, and artificial intelligence.

1.1 Background

Karnataka is one of the culturally richest states in India, known for its diverse and vibrant traditional art forms such as Yakshagana, Dollu Kunitha, Bidriware, Ilkal weaving, Channapatna toys, and many more. These art forms are not only a means of artistic expression but also represent the history, traditions, and identity of the region.

However, in recent years, these traditional art forms have been facing significant challenges. With the rapid growth of digital media and modernization, people are becoming more inclined towards contemporary entertainment and products, leading to a decline in interest in traditional arts. As a result, many artisans and performers struggle to sustain their livelihoods.

One of the major reasons for this decline is the **lack of digital presence and accessibility**. Most artisans operate in rural or semi-urban areas and rely heavily on local markets, exhibitions, or word-of-mouth for promotion. There is no centralized digital platform where users can easily discover these art forms, locate artisans, or participate in cultural events.

Additionally, individuals who are interested in learning these art forms often face difficulties in finding authentic workshops or trainers. Information about cultural events, performances, and training sessions is scattered and not easily accessible to the general public.

Existing platforms mainly focus on either tourism or online shopping, but they do not provide a **holistic ecosystem** that integrates cultural discovery, learning opportunities, artisan connectivity, and event participation in a single application.

Therefore, there is a strong need for a unified digital solution that can bridge the gap between traditional artisans and modern users. The Karunada-Kala Source application is proposed as a solution to address these challenges by providing a centralized, user-friendly, and technology-driven platform for cultural preservation and promotion.

1.2 What the App Does

- Karunada-Kala Source is an Android application that provides a digital platform to explore, learn, and connect with Karnataka's traditional art forms.
- The app allows users to browse different art forms with detailed information, including images, videos, and cultural descriptions. It helps users understand the significance of each art in a simple and engaging way.
- It also includes **Google Maps integration**, which enables users to locate artisans, workshops, and cultural event venues nearby. This helps users move from digital discovery to real-world interaction.
- The application provides an **event system** where users can view upcoming cultural performances and register for them. In addition, a **workshop booking feature** allows users to enroll in training sessions conducted by artisans.
- An **AI-based explanation feature** is included to simplify complex cultural information and make it easy for users to understand.
- The app also supports **user login and favorites**, allowing users to save content for later use and access it even in offline mode.
- Overall, the application works as a **complete cultural discovery platform** that connects users with traditional arts, artisans, and learning opportunities.

2. Problem Statement

Despite Karnataka having a rich cultural heritage, awareness and accessibility of traditional art forms remain low. Through observation and analysis, three core problems stand out:

Problem 1 — Lack of Visibility for Traditional Artists

Many skilled artisans and performers are located in rural or semi-urban areas and have very limited digital presence. They rely mostly on local exhibitions, word-of-mouth, or seasonal events to showcase their work. As a result, their reach remains restricted, and a large audience—especially younger generations—remains unaware of these art forms. This lack of visibility directly affects their income and the survival of these traditions.

Problem 2 — No Centralized Platform for Cultural Discovery

Information about traditional art forms, workshops, and cultural events is scattered across different sources or not available online at all. There is no single platform where a user can explore art forms, find nearby artisans, and check upcoming events together. Even interested users often struggle to discover where to learn or experience these arts, leading to reduced engagement.

Problem 3 — Limited Interaction Between Users and Artisans

There is no direct and structured way for users to connect with artisans for learning, attending workshops, or participating in cultural activities. Communication usually happens through informal channels, making it difficult to access authentic opportunities. This gap reduces both learning opportunities for users and growth opportunities for artisans.

3. Scope of the Application

3.1 In-Scope Features

- The following are part of the app's planned functionality:
- Art explorer displaying Karnataka's traditional art forms with images, videos and cultural descriptions

- Google Maps-based discovery of artisans, workshops and cultural event locations
- Interactive map markers with category icons (artisans, workshops, events) and tap-to-view details
- Event listing system showing upcoming cultural performances with date, time and location
- User registration for events and workshops through the application
- Workshop booking system with date selection and confirmation
- AI-based explanation feature to simplify complex cultural information
- Favorites feature to save art forms, events and artisans for later viewing
- User authentication using Google Sign-In or email/password
- Basic offline support for saved/favorite content
- User profile with activity history (bookings, saved items)
- Clean and visual UI optimized for Android devices
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3.2 Out-of-Scope (Phase 1)

- To keep the first version focused and deliverable, the following are excluded from Phase 1:
- Online payment or e-commerce functionality for buying products
- Advanced AI-based recommendation system
- iOS or cross-platform application support
- Full offline mode with downloadable data
- Integration with official government or tourism department APIs
- Real-time chat system between users and artisans

4. Functional Requirements

- This section describes what the application must do. Each requirement is written from the user's perspective.
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4.1 Authentication

FR-01: Login and Registration

- The app must support login via Google Sign-In (OAuth 2.0)
- Users must also have the option to register using email and password
- On first login, users complete a short onboarding process by entering their name and selecting interests (e.g., art forms they like)
- User sessions must persist across app restarts using secure authentication tokens
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4.2 Art Explorer

FR-02: View and Explore Art Forms

- The app allows users to browse a list of Karnataka's traditional art forms
- Each art form includes:
- Title and category
- Description and cultural history
- Images and video content
- Users can tap any art form to view detailed information
- An "Explain" button allows users to get a simplified explanation using AI
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4.3 Maps Integration

FR-03: Discover Artisans and Events on Map

- Users can view artisans, workshops, and cultural events on Google Maps
- Each location is represented with a category-specific marker
- Tapping a marker displays details such as name, location, and contact information
- Users can navigate to the location using Google Maps directions
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4.4 Event Management

FR-04: View and Register for Cultural Events

- Users can browse upcoming cultural events such as Yakshagana performances and exhibitions
- Each event displays:
- Event name
- Date and time
- Venue location
- Users can register for events directly through the app
- Registered events are stored in the user profile
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4.5 Workshop Booking

FR-05: Book Training Sessions

- Users can book workshops conducted by artisans
- Users select preferred date and time from available slots
- Booking details are stored in the database and confirmation is shown
- Users can view their bookings in their profile

4.6 AI Explanation Feature

FR-06: Simplified Cultural Explanation

- Users can request a simplified explanation of any art form
- The app sends a request to an AI API
- The AI returns a concise and easy-to-understand explanation
- The result is displayed instantly in the app
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4.7 Favorites System

FR-07: Save and Access Content

- Users can mark art forms, events, or artisans as favorites
- Favorite items are stored in the database linked to the user
- Users can access saved items even in offline mode
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4.8 User Profile

FR-08: Manage User Activity

- Users can view their profile details
- The profile includes:
- Saved favorites
- Registered events
- Booked workshops
- Users can update their basic information

5. Non-Functional Requirements

5.1 Performance

Requirement	Target
App cold start (home screen load)	Under 3 seconds on a mid-range Android device
Art Explorer content load	Under 2–3 seconds on 4G network
Map initial load (artisans/events)	Under 3–5 seconds
Event/workshop booking response	Under 3 seconds
AI explanation response time	Under 5 seconds

5.2 Security and Privacy

- All data transfers must use HTTPS / TLS 1.3
- User authentication must be handled securely using Firebase Authentication
- Firebase Security Rules must ensure users can only access their own data (favorites, bookings, profile)
- Sensitive user data (email, profile details) must not be exposed publicly
- AI requests must not store personal user data
- The application must comply with India's Digital Personal Data Protection Act, 2023 (DPDPA)

5.3 Reliability

Core features (Art Explorer, Maps, Events) should work at least 99% of the time

The app should gracefully handle network failures with proper error messages

Cached data (favorites, last viewed content) must be accessible in offline mode

Firebase should ensure data backup and minimal data loss

5.4 Usability

- The interface should follow Material Design 3 guidelines and feel familiar to Android users

- Minimum touch target size of 48dp for all interactive elements
- Color contrast must meet WCAG 2.1 Level AA for accessibility
- App should support English as default, with Kannada as the first regional language

5.5 Code and Maintainability

- The interface should follow Material Design 3 guidelines
- Navigation must be simple and intuitive for all user types
- Minimum touch target size of 48dp for all buttons and interactive elements
- The app should support English as default language, with Kannada as an optional regional language
- UI must be visually rich with images and clear typography

6. Features

6.1 Must-Have Features (Phase 1 — MVP)

These features are essential for the first release of the application. Without these core functionalities, the app will not effectively solve the problem of cultural discovery and artisan connectivity.

The **Art Explorer** is the core feature of the application. It allows users to browse and explore Karnataka's traditional art forms with detailed descriptions, images, and videos. This is fundamental because the primary goal of the app is to increase awareness and understanding of cultural heritage.

The **Maps Integration** feature enables users to locate artisans, workshops, and cultural events using Google Maps. This is important because it connects digital discovery with real-world interaction, allowing users to physically visit and engage with cultural activities.

The **Event Management System** allows users to view upcoming cultural performances and register for them. This feature ensures user engagement and promotes participation in traditional events.

The **Workshop Booking System** allows users to enroll in training sessions conducted by artisans. This feature is critical as it provides a structured way for users to learn traditional art forms directly from experts.

The **AI Explanation Feature** helps users understand complex cultural concepts in simple language. This is important for beginners and ensures that the app is accessible to a wider audience.

The **User Authentication System** enables secure login using Google Sign-In or email/password. This is necessary to store user data such as favorites, bookings, and preferences.

The **Favorites Feature** allows users to save art forms, events, and artisans for future reference. This improves user experience and ensures easy access to preferred content, even in offline mode.

The application also focuses on a **lightweight and efficient UI**, ensuring smooth performance on mid-range Android devices. This is important for accessibility, especially in regions where high-end devices are not commonly used.

6.2 Good-to-Have Features (Phase 2 and Beyond)

These features enhance the functionality of the application but are not required for the initial release. They can be implemented in later phases to improve user experience and scalability.

The **Artisan Marketplace** feature would allow artisans to showcase and sell their products directly through the application, creating an additional source of income and expanding their reach.

A **Ratings and Reviews System** can be introduced to allow users to provide feedback on artisans, workshops, and events. This helps in building trust and improving quality.

The **Recommendation System** can suggest similar art forms to users based on their interests and browsing history, improving personalization.

Push Notifications can be used to alert users about new events, workshops, or updates, increasing engagement.

The **Offline Mode Enhancement** would allow users to access more content without an internet connection, improving usability in low-network areas.

A **Multi-language Support System** can be added to support regional languages like Kannada, Hindi, and others, making the app more inclusive.

An advanced **AI-based chatbot** could be implemented to answer user queries related to art forms and guide them through the app.

7. Technical Implementation

7.1 Technology Stack

The application is developed using **Kotlin** as the primary programming language, which is officially supported for Android development and provides modern, concise, and safe coding practices.

The system follows the **MVVM (Model-View-ViewModel) architecture with a Repository Pattern**, which helps in separating concerns, improving code maintainability, and making the application scalable.

For the user interface, the app uses **Jetpack Compose or XML layouts based on Material Design 3 guidelines**, ensuring a modern, responsive, and user-friendly design.

The **Google Maps Android SDK** is used to display artisan locations, workshops, and cultural events on an interactive map. Navigation support is provided through Google Maps integration for real-world directions.

The application uses **Firebase Authentication** to handle secure user login via Google Sign-In or email/password methods.

For data storage, **Firebase Cloud Firestore** is used to store user profiles, art forms, events, workshop bookings, and favorites. This ensures real-time updates and cloud-based data persistence.

For storing images such as artisan photos or artwork, **Firebase Storage** is used.

AI-based features are implemented using APIs such as **Google Gemini or similar GenAI services**, which generate simplified explanations of art forms.

Push notifications for updates like new events or workshop alerts can be implemented using **Firebase Cloud Messaging (FCM)**.

The project is built using **Gradle with Kotlin DSL**, and version control is managed using **Git and GitHub**.

7.2 AI Explanation Logic

The AI explanation feature is designed to simplify complex cultural information for users.

When a user clicks the “Explain” button on an art form, the application sends a request to the AI API with a prompt containing the art form details. The AI processes the input and generates a simplified explanation in easy-to-understand language.

This response is then displayed in the app, making it easier for beginners or non-experts to understand traditional art forms without confusion.

The generated explanations are not stored permanently but can be cached temporarily to improve performance.

7.3 User Flow Summary

The user opens the application and logs in using Google Sign-In or email/password.

After logging in, the home screen displays featured art forms and a map showing nearby artisans and events.

The user can browse different art forms, view detailed descriptions, and optionally use the AI explanation feature to understand them better.

If the user is interested in an event, they can view its details and register directly through the app.

Similarly, the user can book workshops by selecting available dates and confirming their participation.

The user can also explore the map to locate artisans and navigate to their locations.

Favorite items such as art forms, events, and artisans can be saved for later access.

All user activities such as bookings and saved items are stored and can be viewed in the user profile section.

8. Impact Goals

Karunada-Kala Source is not just an informational application; it is designed to create a meaningful impact on cultural preservation and community development.

Cultural Preservation

The application helps in preserving Karnataka's traditional art forms by digitizing them and making them accessible to a wider audience. It ensures that cultural knowledge is not lost over time.

Economic Empowerment

By providing visibility to artisans and enabling workshop bookings, the app creates new income opportunities for them. It connects artisans directly with users, reducing dependency on intermediaries.

Tourism Promotion

The platform encourages cultural tourism by helping users discover events, workshops, and artisan locations. This can increase footfall in rural areas and boost local economies.

Digital Inclusion

The app empowers rural artisans by giving them a digital presence and connecting them to modern technology, helping bridge the digital divide.

Cultural Awareness

By providing easy access to cultural information and AI-based explanations, the app promotes awareness and appreciation of traditional arts among younger generations.

9. Success Criteria

The success of the application is measured based on how effectively it meets its objectives and delivers core functionalities.

The **data persistence system** must ensure that user data such as favorites, bookings, and profile details are securely stored in Firebase and restored correctly even after app reinstallation.

The **map functionality** must accurately display artisan locations and events, and users should be able to interact with markers without delay.

The **event and workshop booking system** must work seamlessly, allowing users to register and view their bookings without errors.

The **AI explanation feature** must generate clear and meaningful explanations that are easy for users to understand.

The **user interface** must be visually appealing, responsive, and optimized for mid-range Android devices, ensuring smooth performance.

Overall, the app should provide a stable, reliable, and engaging user experience.

10. Risks and How We Plan to Handle Them

One major risk is **low availability of data** such as artisan details and event listings. This can be handled by adding initial demo data and gradually expanding the database through user contributions or partnerships.

Another risk is **API failure or network issues**, which may affect features like maps or AI explanations. This can be mitigated by implementing fallback messages and caching important data locally.

There is also a risk of **fake or incorrect data entries**. This can be addressed by implementing admin verification or moderation systems before displaying user-submitted content.

Performance issues on low-end devices can impact user experience. To handle this, the app will be optimized with lightweight UI design and efficient background processing.

Lastly, **low initial user adoption** can make the platform less useful. This can be improved by promoting the app through colleges, cultural organizations, and social media platforms.

End of Document

Karunada-Kala Source (National Pride) | Android App Development using GenAI